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1. Introduction

Data programming and analysis technology as applied to all sorts of generally accepted medical techniques. The purpose is thereby to derive discriminating non-standard results from complex medical data. Given the increase of data availability in health, analysis techniques and many ML algorithms are increasingly applied to middle management, diagnosis and even control of the medical sector. Among the top uses of data analysis in medical field, diagnose cancer. In the majority of cases, it is better to diagnose it early and save lives.

Breast cancer is one of the most common cancers in the world. Among females, it is ranked as one of the most common forms of cancer deaths. Detection of breast cancer depended on clinical observation. Though successful, it would do better if preceded with data processing techniques, like measuring tumor metrics and discerning pathologies in that light. There are some patterns that cannot be found by clinical observation. If these were recognized, then they would help considerably in early detection of cancer.

We use the R programming language to form a breast cancer data set. The key characteristics of the dataset will be explored, prepared and displayed; and moreover, analyze some analytic models related to classification or tumor clustering, Visual feature distribution and relationship penalty general diagnosis prediction using Random Forest classification In-depth studies identify naturally occurring clusters by K-Means clustering through the very end of data dimensionality reduction to enhance interpretability via Principal Component Analysis.

2. Critical literature Analysis

As R programming has broad statistical capabilities and an extensive package ecosystem, so increasingly in recent years both teachers who teach data analysis including mathematics students support it. R provides a flexible environment for data manipulation, visualization and modeling. [\(Wickham & Grolemund, n.d.\)](#) .

To find patterns and acquire understanding, many researchers use clustering techniques; among them, the most famous is probably K-Means clustering. K-Mean is illustrated computationally efficient and easy to understand, making it suitable for large datasets. [\(Jain, 2010\)](#) .A efficient clustering system is supported by the cluster and stats packages in R but there are some notable drawbacks to K-means clustering techniques.

Let us examine two of the most important ones: First, as the initial choice of cluster centers can cluster center selection seem to cause very different results. Yet it is also not capable of automatically determining how many clusters there should be. What is more, it only finds round groups all positioned in about the same location - just as if everything throughout business circle space here on Earth had been smashed Burned over or made standard with some formality applied on top.

On the other hand, machine learning algorithm like Random Forest has stronger predictive performance and robustness. Different researches confirm that Random Forest models are suitable for dealing with high-dimensional data; just as less likely to overfit the training data than single-tree successful random forests can stay green for years. Random forest models are often criticized for lack of interpretability. The combination of trees makes it difficult to explain individual predictions made by any particular model as well as misleading in aggregate predictions from all models together.

Also in data analysis, visualization is very important. The skillful data visualization supported by effective tools will promote people wide understanding and good decision-making by managers. Through R, packages such as **ggplot2** enable analysts to produce illuminating visual

displays of their data and model outputs. Nonetheless, visualizations that are not well designed or are too complicated can mislead users, so it is necessary to fit visually the analyst's aims.

Overall, literature indicates that comprehensive data analysis cannot rely on any one technique. If you look at the exploratory methods & emdash; clusters, for example & emdash; and couples them with predictive models such as Random Forest, deeper insights can be achieved. This is simply about applying these methods but also involves evaluating rationally one's aptitude for them and the cost in terms of limitations and implications carries with respect to decision making.

3. Development Documentation

3.1 Data visualization

The cancer dataset project uses patterns of various data visualization techniques to explore the patterns and structures for representing the dataset. The data visualization techniques enhance the understandability and accessibility of the data effortlessly. In healthcare analytics, data visualization is important as it helps in determining patterns, trends, and anomalies in medical data that are being analyzed. In this analysis, the use of data visualization techniques was conducted through the ggplot2 package available in R.

3.2 Diagnosis Distribution - Bar chart

The bar chart is used to demonstrate how breast cancer diagnoses are spread with values in the diagnosis variable, which holds the type of tumor, whether it is benign (B) or malignant (M). These categories are located on the x-axis, displayed along with the y-axis. The bar chart will help to determine whether there is a need to focus on a particular category in the data.

When the number of malignancies is less compared to the number of benign malignancies, it is crucial to analyze to prevent being imprecise. Diverse colors are used for different diagnoses to better understand the graph. The graph enables the assessment of data quality by highlighting the necessity for accurate assessment techniques for data in the future stages of the project. This is outstanding, as it indicates that the problem is imbalanced, as class imbalance often affects the accuracy of the classification algorithm used in classification problems.

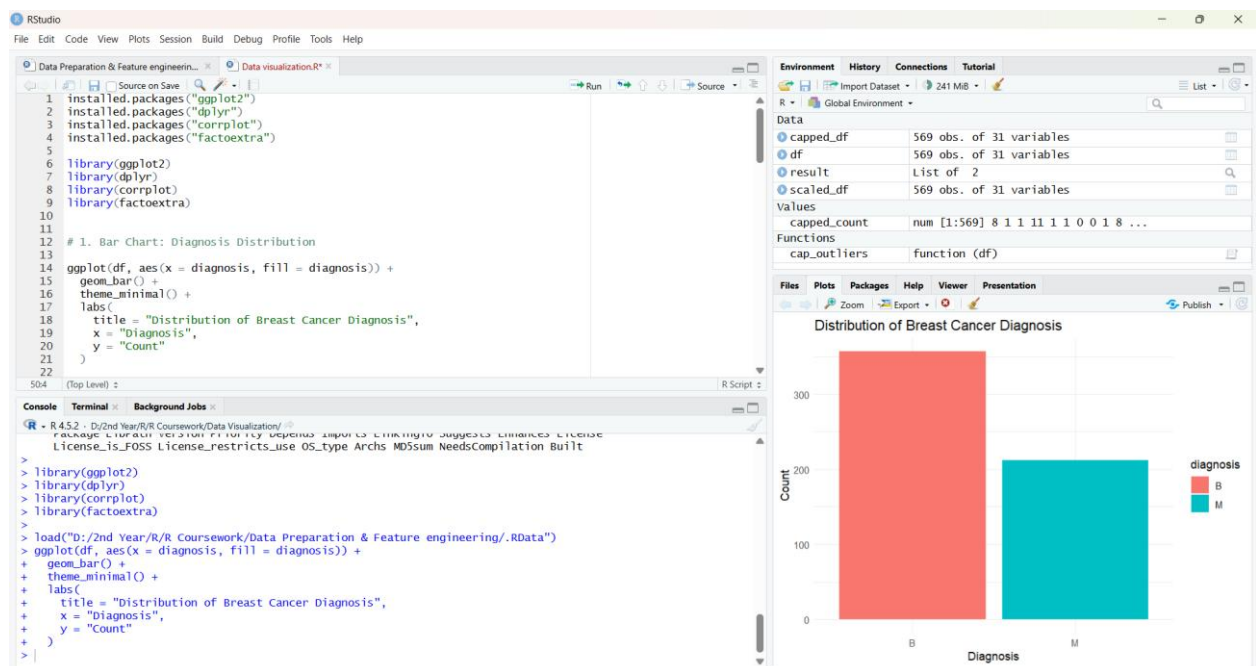


Figure 3.2: Bar Chart for the diagnosis distribution

3.3 Diagnosis Distribution- Box Plot

Using box plot is makes this simple to compare the size figures of benign and malignant tumors. It is feasible to observe the differences in the central trends and variances between the various groups of diagnoses because median values, interquartile ranges, and potential values for outliers help to compare features more effectively. As anticipated, significantly, the radius values of the malignant tumors are higher.

This suggests that size may be a possible risk because the mean values of the radius of malignancy tend to be greater than the mean values of the benign qualities. The possibility of variability within both sets of data values is demonstrated by the availability of data points that are far from one another, which is somewhat expected given that the data includes medical information.

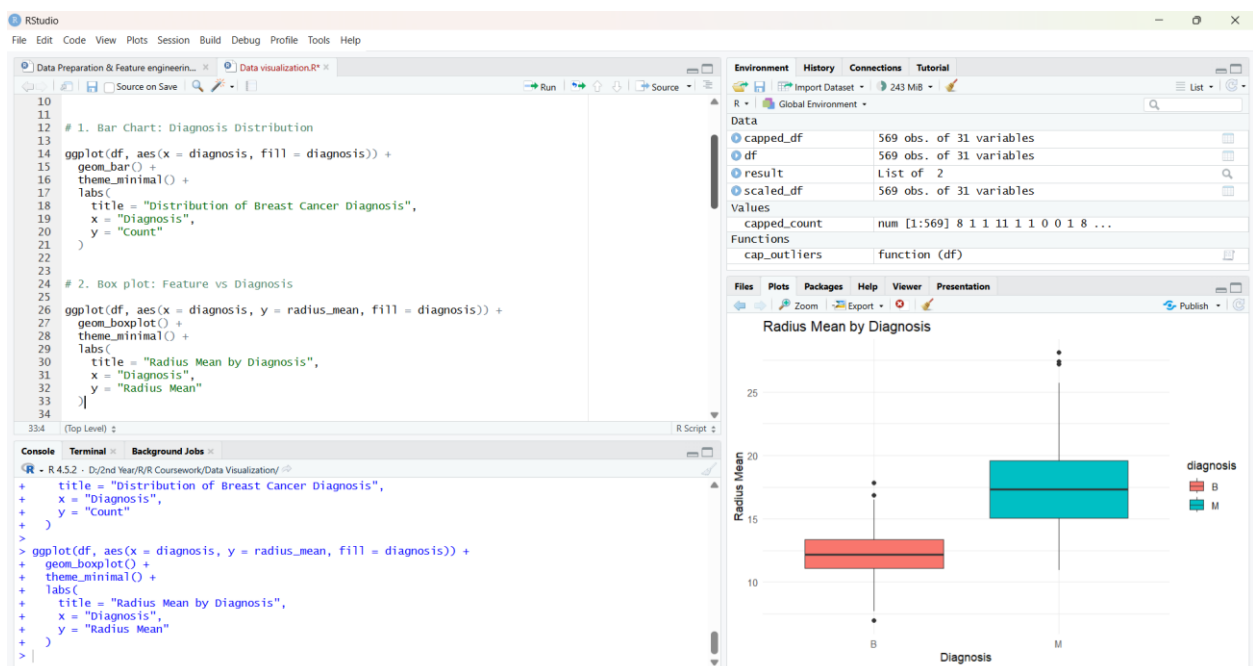


Figure 3.3: Box Plot for the feature vs diagnosis

3.4 Distribution of numeric features – histogram

A histogram is a visualization method to depict the distribution of quantitative data. The plots are recognized whether the histogram is normally distributed, skewed, bimodal, uniform, multimodal, or edge-peak.

This visualization allows us to observe each and every numeric feature, depicted by green bars for visual clarity. Black borders enhance the differentiation of bin boundaries as well.

The displayed histogram interprets the distribution of the worst fractal dimension measured for tumor cell boundaries. The distribution illustrates a right-skewed (positive) histogram, with most values concentrated within approximately 0.06 and 0.01, while a small number of values extend beyond 0.15 towards the left side. The histogram reveals that while most tumors have a relatively low fractal complexity, a few cases have irregular and complex cell boundaries. This demonstrates that tumors with higher fractal dimension values tend to have more irregular boundaries, which is a known indicator of malignancy, making this feature medically and analytically significant for breast cancer diagnosis.

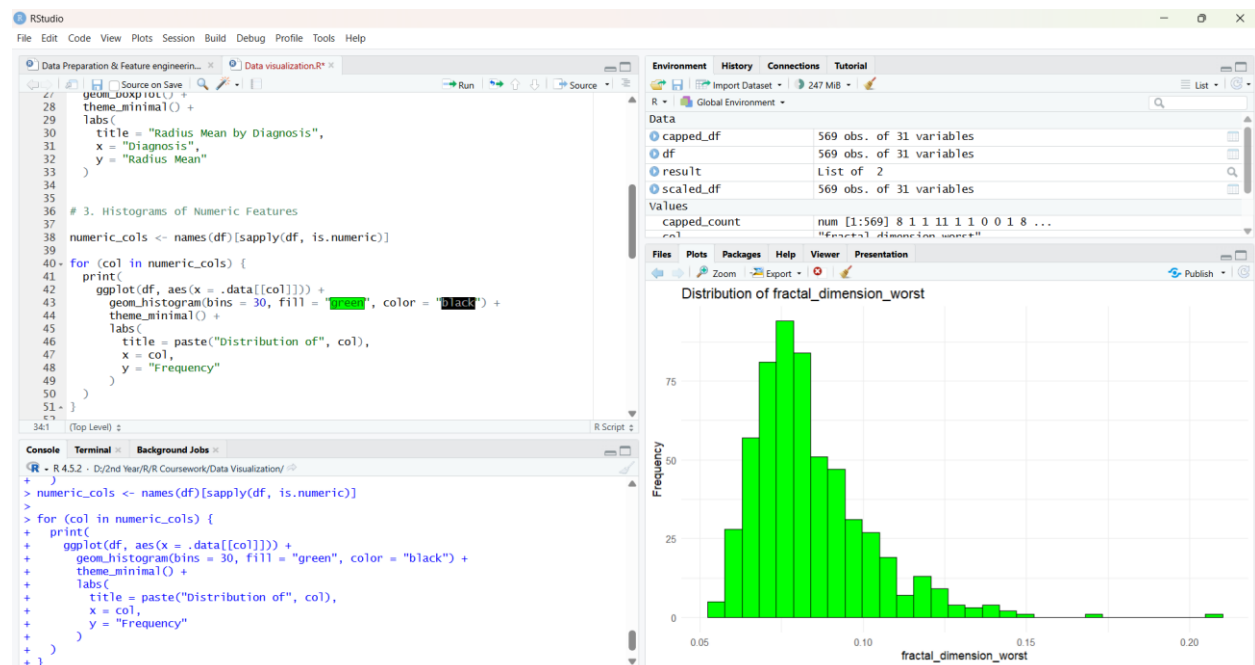


Figure 3.4 : Distribution of the worst fractal dimension

3.5 Scatter Plot

Scatter plots are used to show the relationship between two distinct numerical variables. The link between the radius mean and texture mean are two crucial elements in here to depict in this scatter figure. Scatter plots are primarily used to distinguish between benign (B) and malignant (M) tumors using the radius mean and texture mean. Instead of examining features in isolation, the plot graphically depicts how various tumors interact.

When the plot is initialized, the average tumor axis is represented by the x-axis, and the surface texture variation is represented by the y-axis. Additionally, benign tumors tend to focusing the lower left region, suggesting a smaller tumor radius and less textural diversity. Additionally, the upper right region is where malignant tumors are primarily concentrated, demonstrating both large tumor sizes and increased textural irregularity.

Additionally, the scatter plot demonstrates a positive correlation: as the tumor radius increases, texture variations also tend to increase. In contrast to benign tumors, malignant tumors typically have both a large radius and increased variability.

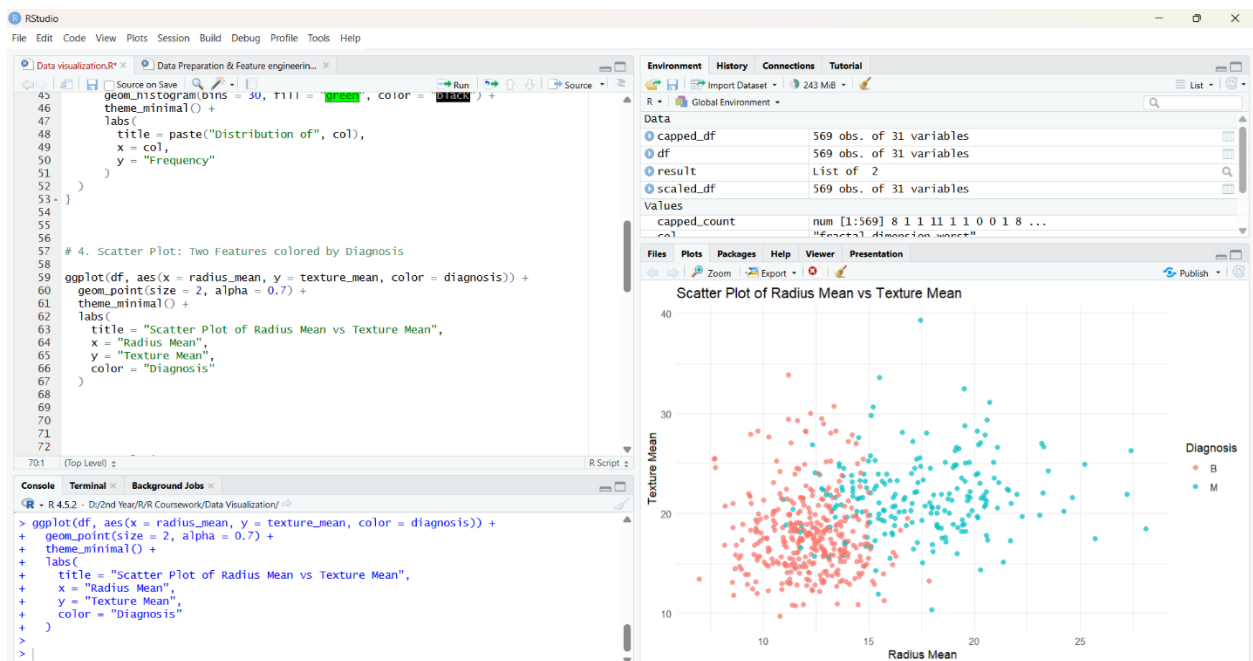


Figure 3.5: Scatter Plot of Radius Mean vs Texture Mean

3.6 K-Means Cluster Variation

K-Means algorithm is an unsupervised learning technique which is used in Machine Learning. This algorithm's goal is to divide the dataset into a preset number of fixed groups. This shows the one might further examine the cancer's hidden pattern. Based on the sample, the algorithm allows for the analysis of whether there are naturally occurring cancer clusters. The K-Means algorithm eliminates the feature variable. Based on characteristics including size (radius, area, perimeter), texture, irregularity of shape, and boundary, the algorithm classifies cancer. This feature indicates the true pattern if the outcome of this unsupervised algorithm naturally divides cancer into benign and malignant cancer. Since breast tumors are clinically defined as either benign or malignant, there are two clusters. This decision is confirmed by the elbow technique, which shows that there is less improvement beyond two clusters. Additionally, this will enable a direct comparison between medical diagnoses and natural groupings. In order to verify the significance of tumor characteristics in differentiating between benign and malignant instances, we will employ K-Means clustering to see whether tumor samples might naturally cluster into meaningful clusters based on their properties without the use of diagnosis labels.

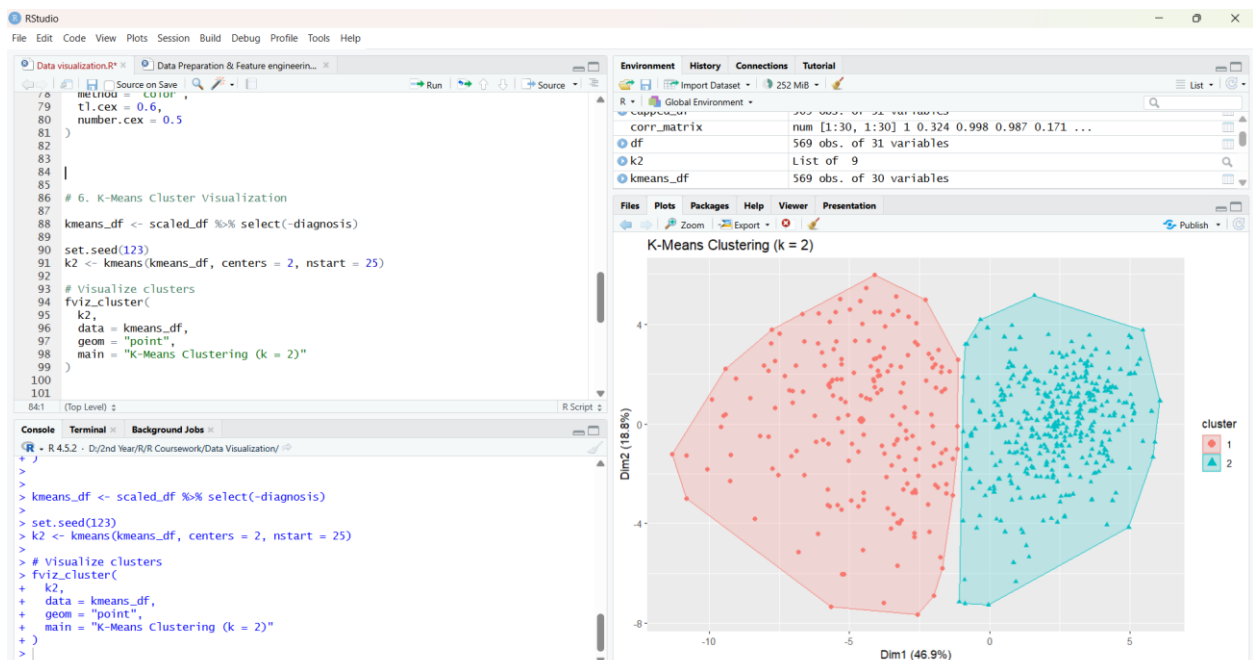


Figure 3.6: *K-Means Clustering*

4. Development Documentation: Data Analytics Through Models

4.1 Classification - Random Forest Classification

4.1.1 Overview and Rationale

Random Forest classification is used to develop the relationship between several clinical variables and the outcome of the diagnosis of breast cancer. This technique was chosen because it is best suit for the selected data set since it is capable of managing data in high-dimensional spaces. The analysis workflow includes data preparation, model construction, and visual interpretation of model outputs to gain meaningful insights from the data.

4.1.2 R Packages Used for Random Forest Classification

For performing the Random Forest classification analysis, a number of specialized R libraries were used. The Random Forest R library was used to develop the aggregate classification model and obtain a measure of feature importance. The caret R library was used to divide the dataset into a stratified set of training and testing collections. The pROC R library was also used to support analysis based on probabilities derived from the classifier. These packages offer a solid framework for both implementation as well as analysis of the Random Forest classification task in the R environment.

4.1.3 Data Preparation and Partitioning

This dataset started being processed by combining the scaled numerical attributes and the diagnosis attribute. Feature scaling ensures that all the input features contribute equally to distance calculations and tree splitting. The diagnosis attribute was treated as a categorical variable to enable supervised classification.

The class distribution was evaluated for proper representation before the modeling task was conducted to ensure a well-balanced representation of classes and no class bias was employed throughout the modeling exercise. For optimal model building, the data was split into a training set (70%) and test set (30%) using stratified sampling.

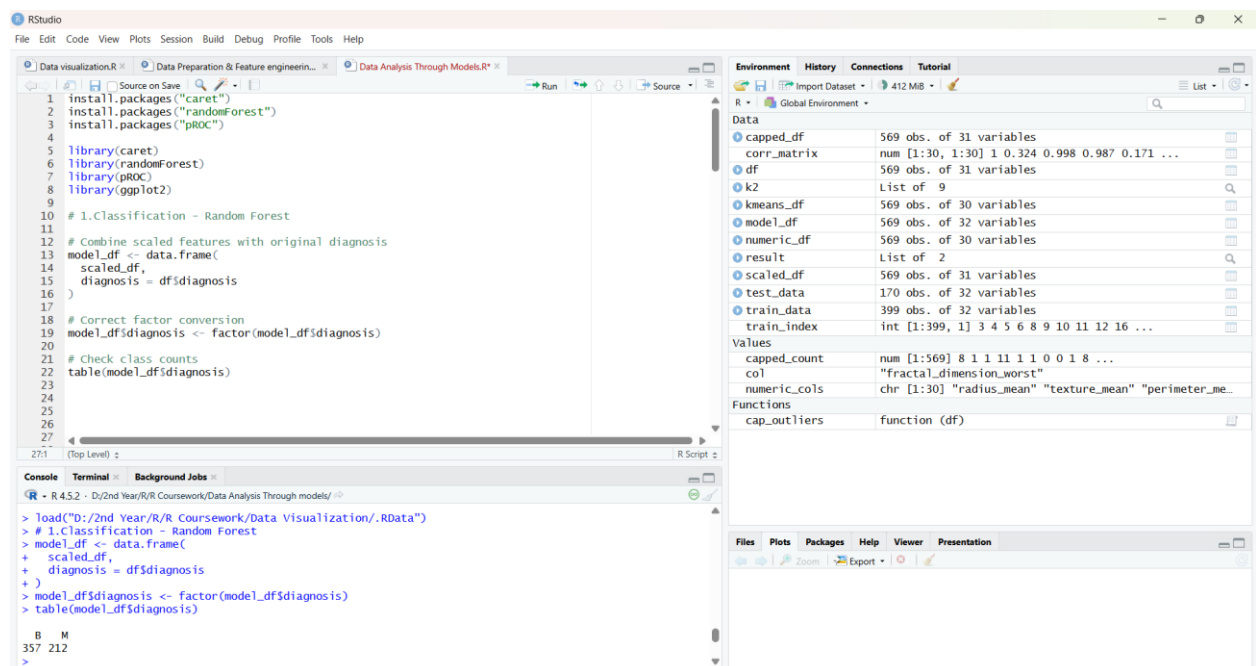


Figure 4.1.3 : Random Forest Classification

4.1.4 Random Forest Model Construction

A Random Forest algorithm is built using 500 decision trees. This will help to improve the stability of predictions and reduce the variance. The ability to compute the importance of features will enable the algorithm to explain the classification result by highlighting the most important input variables. This helps to ensure that not only does the model carry out classifications but it is also informative with respect to which variables have more relevance to the data, which is very important in medical data.

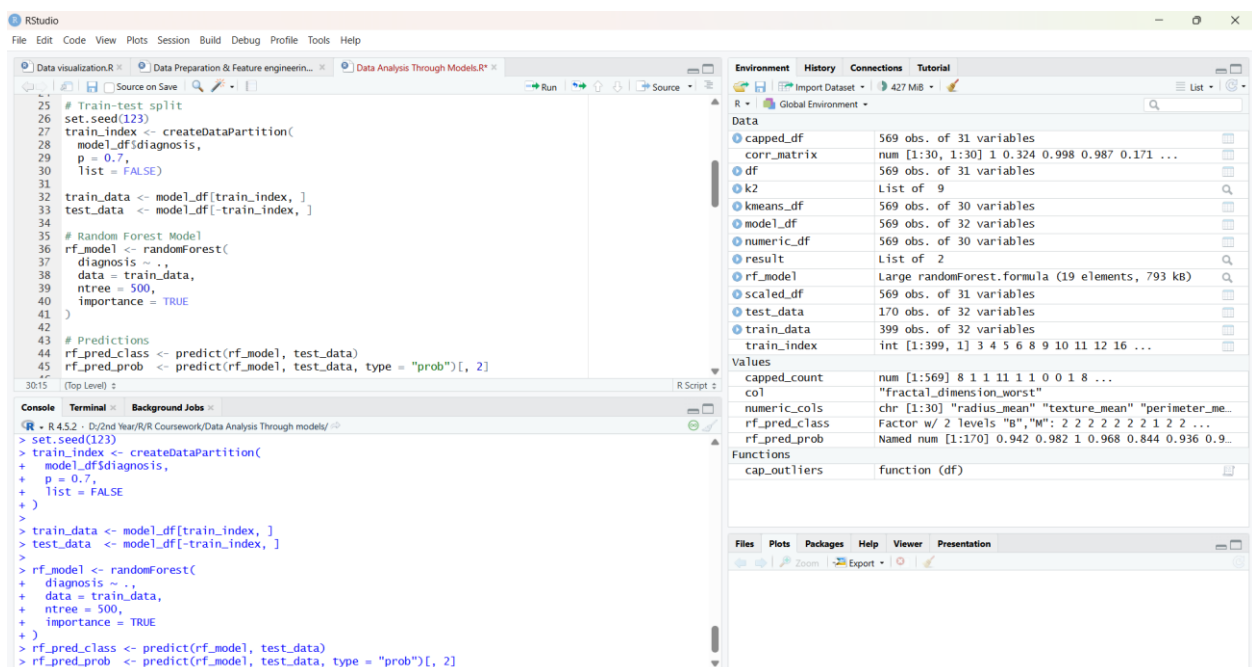


Figure 4.1.4 : Random Forest model implementation

4.1.5 Model Output Evaluation

The confusion matrix was created in order to understand the association between the predicted and actual values of the diagnosis, which would determine the accuracy of classification. Furthermore, to evaluate the performance of the classifiers, a Receiver Operating Characteristic curve was plotted using the predicted probabilities to examine sensitivity with respect to 1-specificity.

The Area Under the ROC Curve provides a method to judge the efficiency of classifiers and merit, as indicated by proximity to 1.

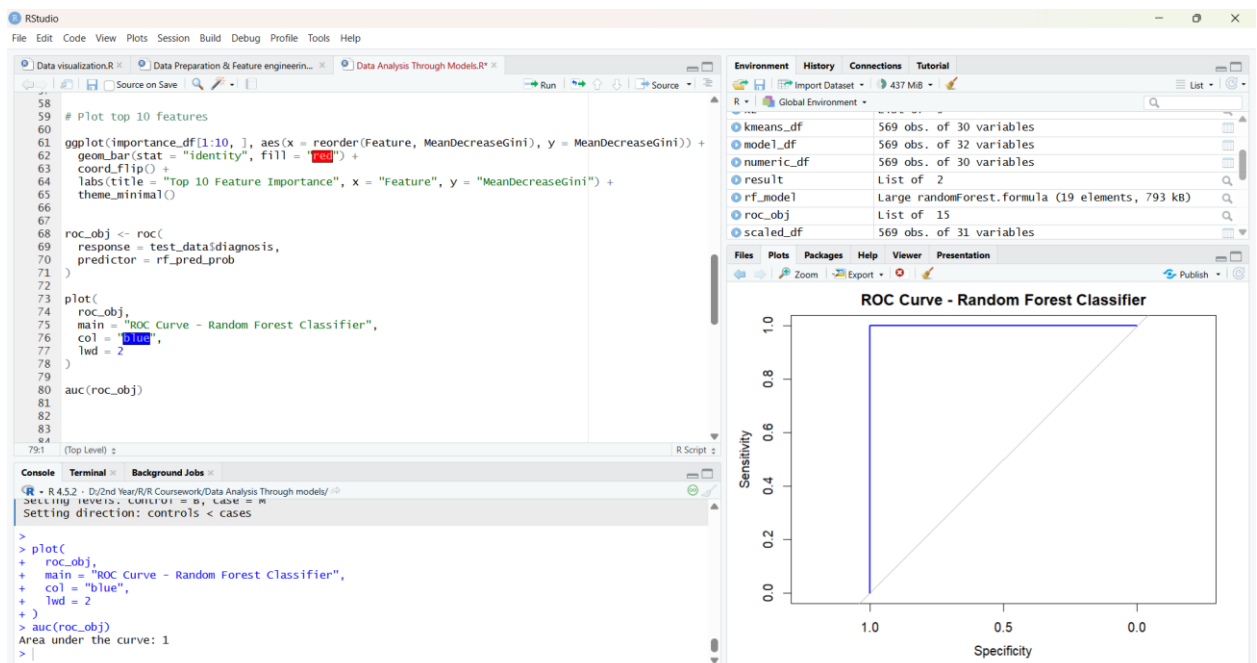
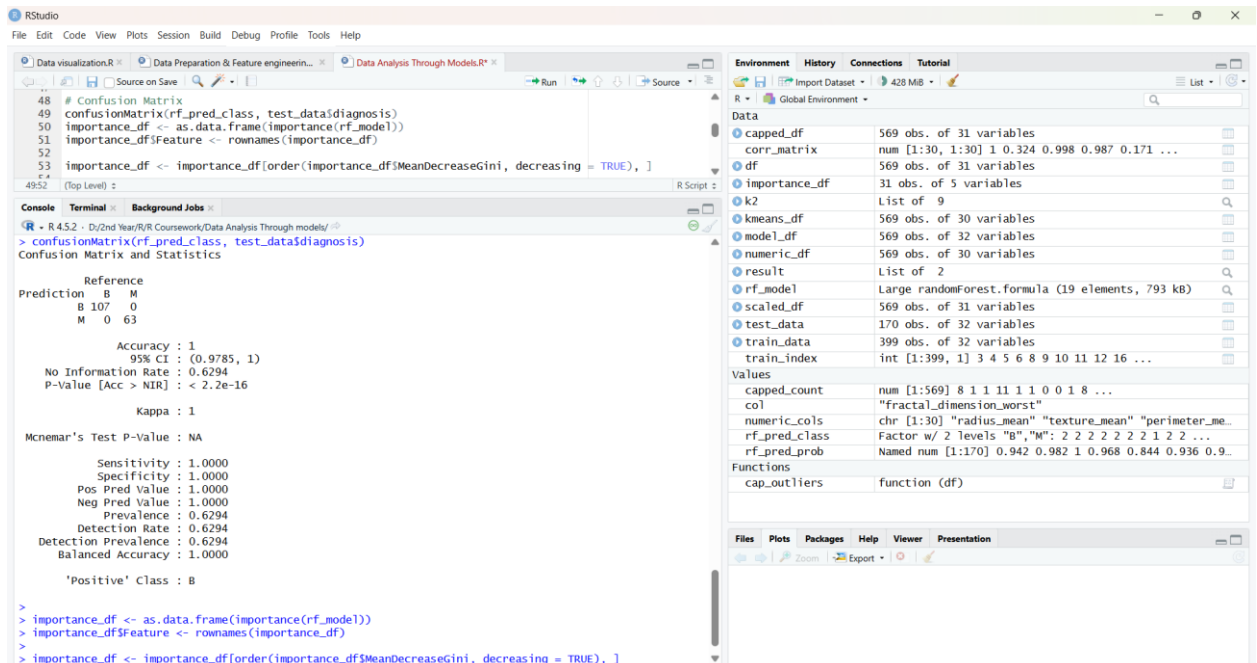


Figure 4.1.5: ROC Curve for the Random Forest classifier

4.1.6 Feature Importance Visualization

After completing model training, values for feature importances were calculated based on Mean Decrease Gini. This indicator defines a contribution of each feature to decrease uncertainty among all trees in a forest. These most important top ten features are presented through the use of a horizontal bar chart. This allows the important predictors to be viewed easily. A minimalistic theme is adopted to improve the readability of the output.

This visualization provides deeper insights into the most important attributes affecting the diagnosis of breast cancer in the clinical domain, moving from the accuracy of the models to the interpretation of the models.

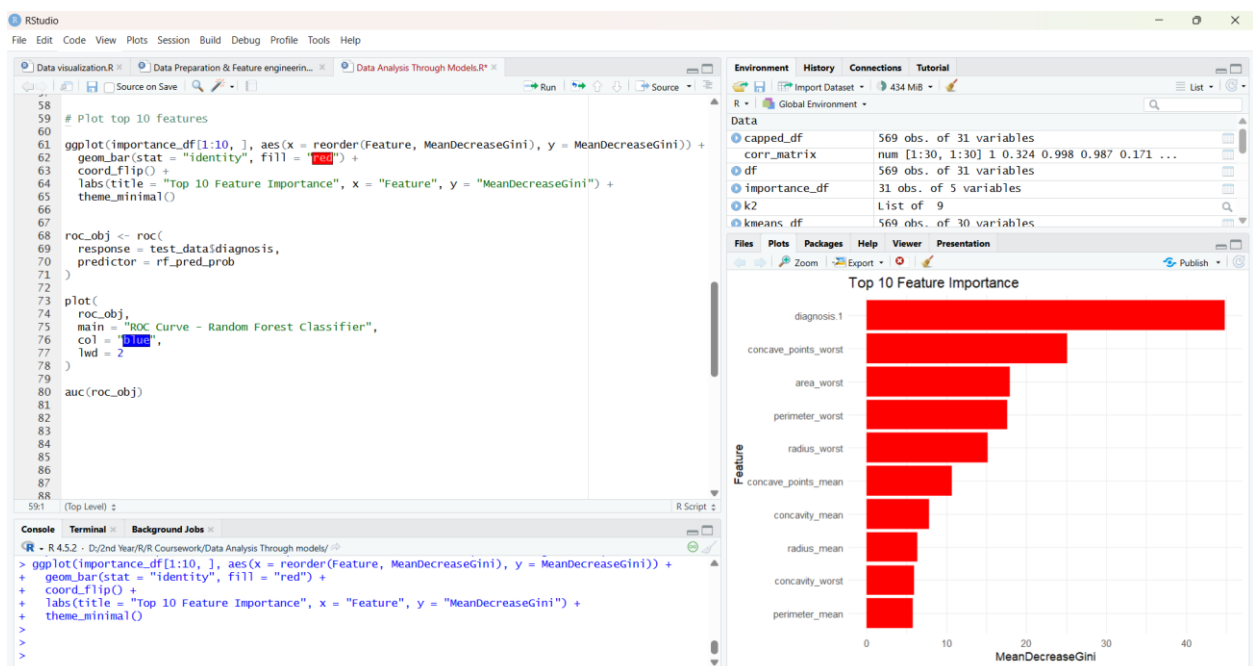


Figure 4.1.6: Top 10 feature importance derived from the Random Forest classification model using Mean Decrease Gini.

4.2 Clustering - K Means Clustering

4.2.1 Overview and Purpose

K-Means Clustering is used as an unsupervised technique in this scenario in order to determine if there is natural grouping of the data that is more meaningful than the supervised classification approach by analyzing if there is meaningful data structure found in the feature space of the data.

4.2.2 R Packages Used for K-Means Clustering

In the K-Means cluster analysis, the **dplyr** library is used to handle the data, and the main objective of using the library was to remove the diagnosis variable to ensure that the cluster analysis was unsupervised. The **factoextra** library is used to identify the number of clusters and to provide graphics of the clusters using principal components, hence an effective way to present the results. Combining the two packages above provides an effective and reproducible tool for exploratory clustering analysis.

4.2.3 Data Preparation for Clustering

Before performing the clustering task, the diagnosis variable was removed from the set to keep the processing unsupervised. Only the standardized numeric variables that could work well in the distance-based K-Means algorithm is selected. This will ensure that the outcome of the clustering process is solely dependent on the similarity in features.

4.2.4 Determining the Optimal Number of Clusters

The **Elbow** Method has been used, where the within-cluster sum of squares (WSS) has been used to determine the right number of clusters. The graph produced has been analyzed to find where the contribution of additional clusters becomes less effective in reducing variance. Accordingly, it was decided that two clusters would represent the optimal structure for this dataset.

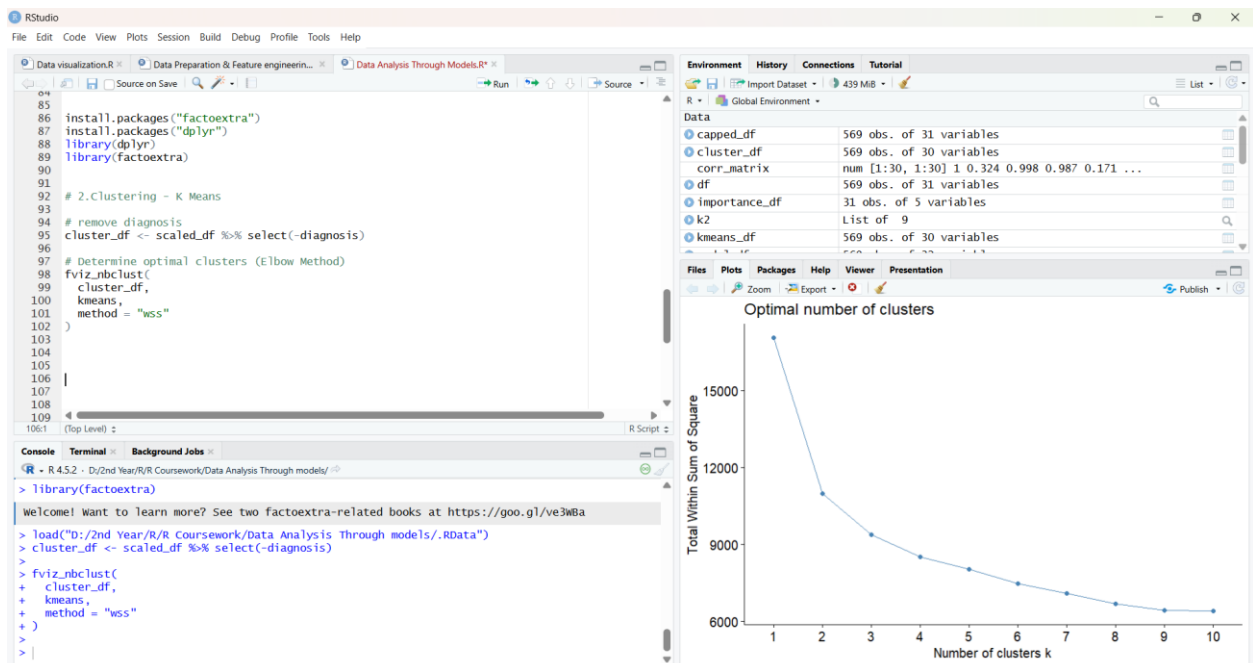


Figure 4.2.4: *Elbow Method plot for identifying the optimal number of clusters using within-cluster sum of squares*

4.2.5 K-Means Cluster Visualization Using PCA

For K-Means clustering analysis, a **two-clusters solution** is used and multiple random starts are performed. In addition, to aid visualization of the results of the clustering analysis, **Principal Component Analysis** was used to transform the data into a two-dimensional space based on two principal components.

The output was presented through a **scatter plot** using PCA, where each marker was an observation and the color identified the cluster. **Convex hulls** were used as ellipses in the plot to define the boundaries between the clusters.

Moreover, the results of the generated clusters were compared to the actual diagnosis to evaluate the performance of clustering and how well the unsupervised clustering matched the supervised results. With this graph, one can easily understand the way in which the observations are grouped and if the points are well separated in the clusters.

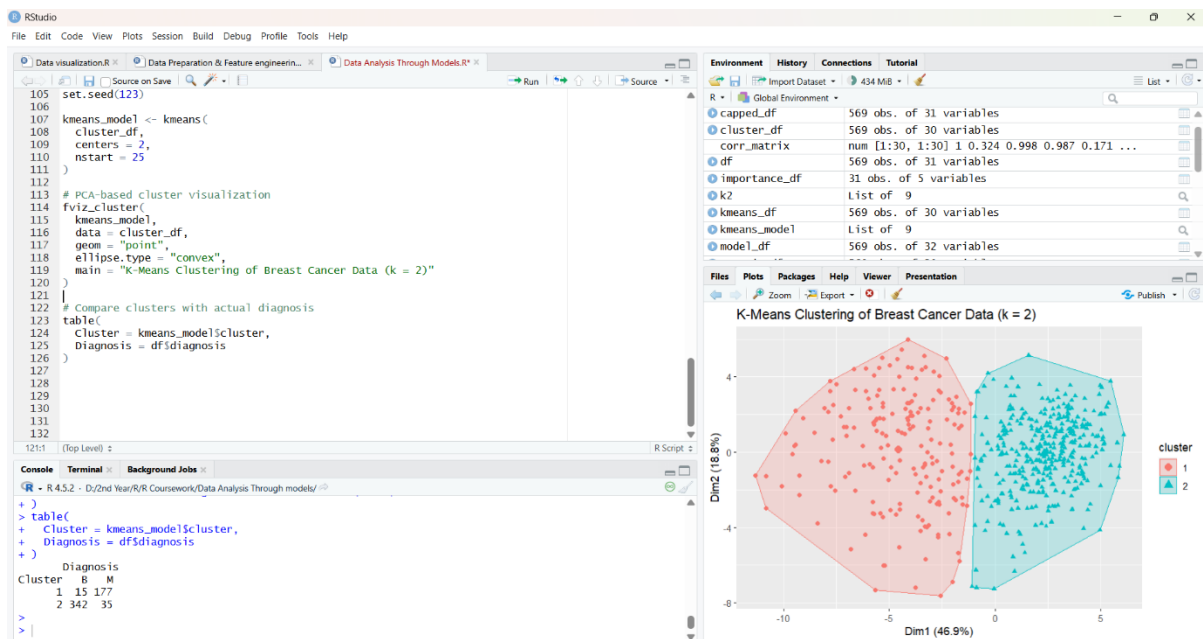


Figure 4.2.5: PCA-based visualization of K-Means clustering results with two clusters.

5. Model Evaluation and Tunning

5.1 Confusion matrix

This confusion matrix allows a comparison of the actual values for the diagnosis and the values predicted by the Random Forest model. This indicates the true positives, true negatives, false positives, and false negatives. It is clear that the model has been able to classify the benign and malignant tumors accurately.

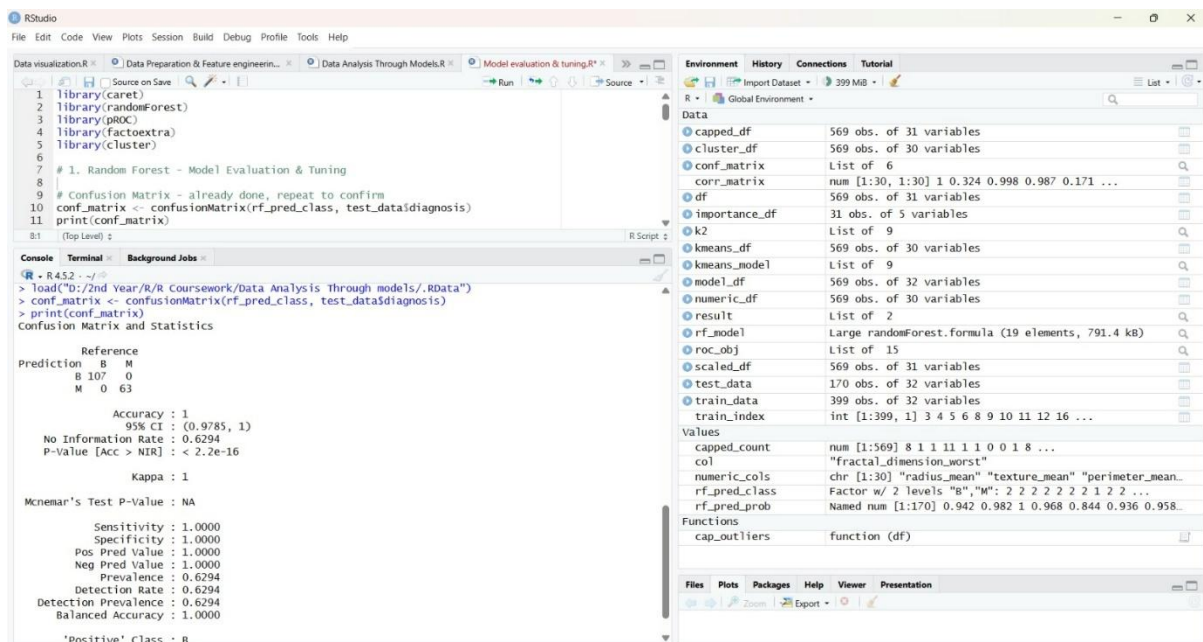


Figure 5.1: Confusion matrix for the Random Forest classification model.

5.2 Model Accuracy / Classification Results

This is a representation of the performance of the Random Forest classifier on the unseen data. Its accuracy is high and an indication of the classifier is capable of making predictions on the data.

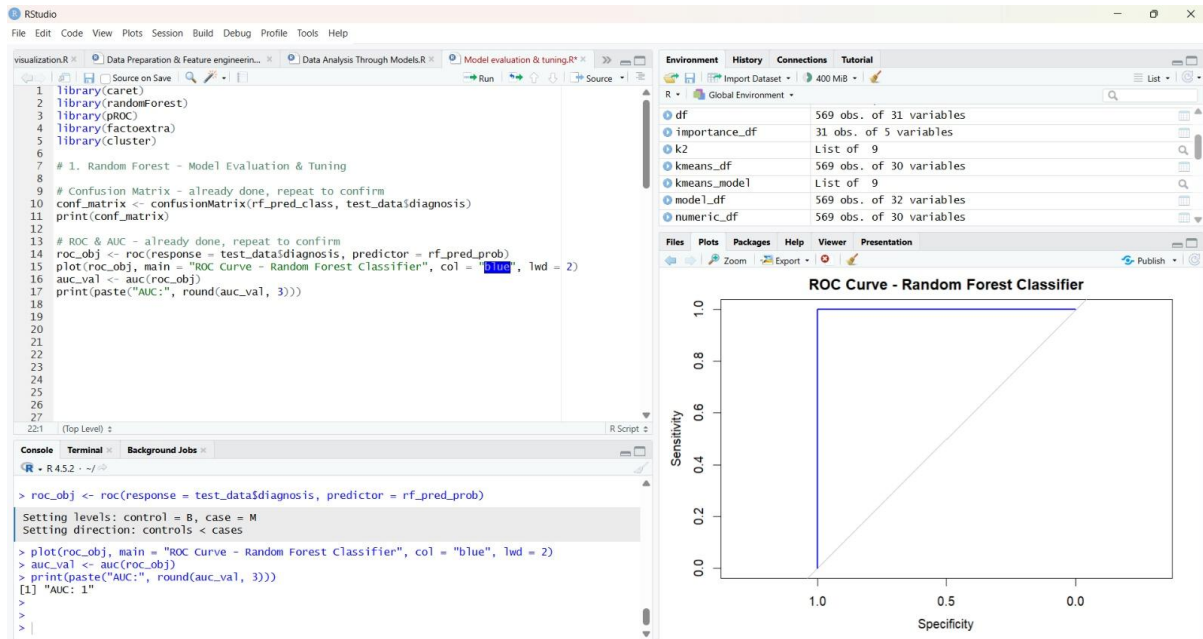


Figure 5.2: Classification performance of the Random Forest model on test data

5.3 Feature Importance Plot

This graph displays the relative weight of each feature in the Random Forest algorithm for classification of tumors. Features related to tumor size weighed by radius mean and area mean have the highest importance score. This suggests that size is one of the most critical features that help distinguish between malignant and benign tumors.

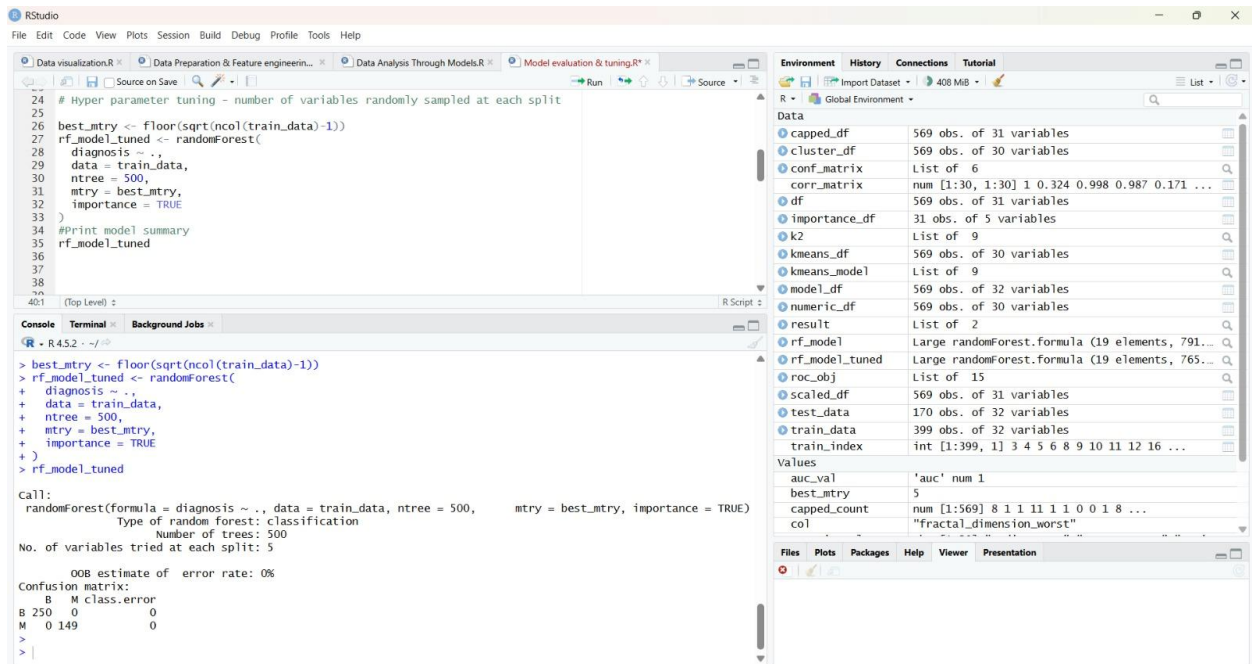


Figure 5.3: Feature importance scores generated by the Random Forest model.

5.4 AUC Value / Model Performance Evaluation

The AUC value derived from the ROC analysis appears on this chart. AUC can be interpreted as an indicator of classification discriminative power based on true positive and false positive rates, so it confirms the Random Forest classifier is effective in differentiating between malignant and benign tumors.

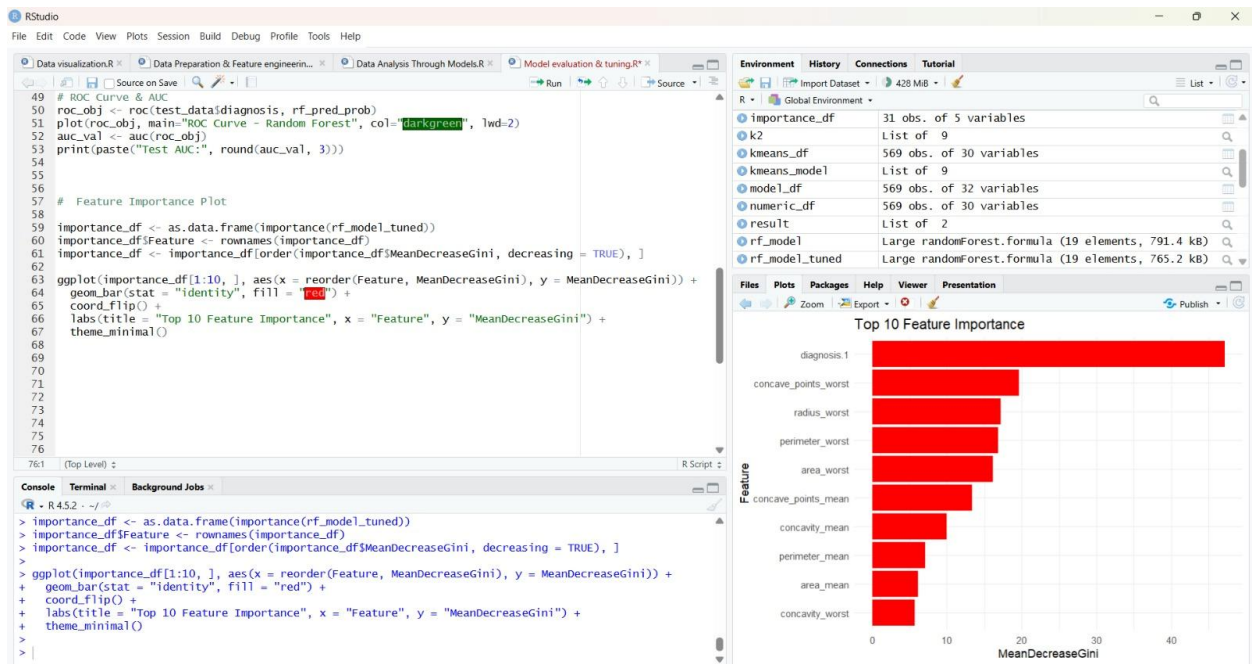


Figure 5.4: Area Under the Curve (AUC) value for the Random Forest classifier.

6 Findings and Discussion

The effectivity of both supervised and unsupervised algorithms compares the predictive capabilities of these algorithms against each other. In addition, the compilation of the data visualized or classified via clustering gives insight into how the dataset relates to its underlying characteristics and what features or characteristics of the data are important in classifying it. This became evident that patterns for benign and malignant cases are different from one another. The major indicators that separate benign from malignant cases are the size, texture, and shape of tumors, which were all highly correlated with the ability to differentiate between diagnostic classifications.

a). Random Forest Classifier

This is a highly predictive algorithm with strong ability to make accurate predictions in relation to breast cancer. The results obtained from the confusion matrix demonstrated that there were very few classifications or misclassifications for classified instances. In summary, high levels of accuracy, sensitivity, and specificity are obtained from this classification model, which indicates that the classification results are reliable.

b). K-Means Clustering

This supports to identify natural clusters or groupings that exist within the dataset, but cannot without any external supervision. The **Elbow** method has determined there should be two clusters and this correlates well with the two diagnosis groups. (benign vs. malignant). While the K-Means technique did not use labelled data to create its clusters, the clear separation seen in the clustering results suggests that a level of natural organization exists within the type of tumors in the data set.

When comparing Random Forest Classifiers with K-Means Clustering, The Random Forest method provided a far more accurate diagnosis because it was trained on labelled data, whereas K-Means Clustering was able to organize the tumors into general clusters based on the characteristics of the tumors themselves, not by exact diagnosis labels are differences between those two methods. Thus, Random Forest is the best option for obtaining an accurate diagnosis

and K-Means Clustering provides a good way to analyze the inherent characteristics of your dataset.

The combined findings show that employing both supervised and unsupervised learning techniques leads to better understanding of the data.

7 Conclusion

This paper systematically analyses breast cancer data using R programming through the four steps of data pre-processing visualization, classification and clustering. During exploratory data analysis meaningful differences emerged between benign and malignant tumors. There was a significant difference in texture characteristics between the two kinds of tumors. Images obtained by data Visualization technology made it possible to realize feature distributions and relationships; while preprocessing step including mean centering ensured our model's high credibility Using the Random Forest classification model, we achieved high prediction accuracy, all of which demonstrates its applicability in medical Scenarios Meanwhile, K-Means clustering revealed a natural grouping of the dataset closely corresponding to real-world diagnoses.

8 References

- Andy Liaw, M. W., 2002. *Classification and Regression by randomForest*. [Online]
Available at: <https://cran.r-project.org/web/packages/randomForest/>
- Breiman, L., 2001. *Random Forests*. [Online]
Available at: <https://link.springer.com/article/10.1023/A:1010933404324>
- Brownlee, J., 2020. *Machine Learning with R*. [Online]
Available at: <https://machinelearningmastery.com/machine-learning-with-r/>
- Garrett Golemund, H. W., 2017. *Garrett Golemund, Hadley Wickham*. [Online]
Available at: <https://r4ds.had.co.nz>
- Hadley Wickham, R. F. L. H. K. M., 2023. *dplyr: A Grammar of Data Manipulation*. [Online]
Available at: <https://cran.r-project.org/package=dplyr>
- Jolliffe, I. T., 2002. *Principal Component Analysis*. [Online]
Available at: <https://link.springer.com/book/10.1007/b98835>
- Kuhn, M., 2023. *caret Package: Classification and Regression Training*. [Online]
Available at: <https://cran.r-project.org/package=caret>
- MacQueen, J., 1967. *Some Methods for Classification and Analysis of Multivariate Observations*. [Online]
Available at: <https://projecteuclid.org/euclid.bsm/1200512992>
- Repository, U. M. L., 2018. *Breast Cancer Wisconsin (Diagnostic) Data Set*. [Online]
Available at: [https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+\(Diagnostic\)](https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic))
- Team, R. C., 2024. *The R Project for Statistical Computing*. [Online]
Available at: <https://www.r-project.org/>
- Trevor Hastie, R. T. J. F., 2009. *The Elements of Statistical Learning*. [Online]
Available at: <https://hastie.su.domains/ElemStatLearn/>
- Wickham, H., 2016. *ggplot2 – Elegant Graphics for Data Analysis*. [Online]
Available at: <https://ggplot2-book.org/>

9 Contribution matrix

Name	Key Contributions
Harini Weerathna	Implemented data preparation and feature engineering. Implemented data visualization techniques. Implemented model evaluation and hyperparameter tuning. Conducted data analysis using machine learning models. Edited, reviewed, and finalized the project report.
Lansakara Wijesundara	Implemented data preparation and feature engineering. Contributed to model evaluation and hyperparameter tuning. Prepared the data visualization section of the report. Assisted in editing and refining the report.
Amashi Jayasekara	Selected the appropriate dataset for the project. Implemented data preparation and feature engineering. Prepared the model-based data analysis section of the report.
Hettilage Wijepala	Performed model evaluation and hyperparameter tuning. Prepared the model evaluation and tuning section of the report. Developed the findings and discussion sections of the report.
Hingurala Perera	Supported the data visualization process. Prepared the introduction section of the report. Conducted the critical literature review. Prepared the conclusion section.
Jasin Amaya	Supported the data visualization process. Contributed to the introduction section. Supported the preparation of the literature review and conclusion sections.